

Hannes Vandecasteele, PhD

Machine Learning Engineer • Agentic AI • Scientific Computing

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Machine Learning Engineer specializing in **Agentic AI for research, Scientific Machine Learning, and Computational Science**.

PhD in Applied Mathematics, former Johns Hopkins researcher and Meta engineer with experience building autonomous ML systems, GPU-accelerated scientific software, and production AI platforms.

Published in *Nature Machine Intelligence* and recipient of an **EB-1A Green Card for Extraordinary Ability**.

Experience

Research Engineer

2025–Present

Campbell & Company — Quantitative Hedge Fund

- Developed machine learning infrastructure and AI models supporting quantitative investment research.
- Reduced computational workflows from **hours to seconds** through GPU acceleration and algorithmic optimization.
- Built firm-wide research tools and contributed to AI initiatives spanning multiple investment teams.

Scientific Machine Learning Researcher

2024–2025

Johns Hopkins University

- Developed Scientific Machine Learning algorithms for computational physics and nonlinear dynamical systems.
- Designed Neural Operator architectures accelerating system-level scientific computations.
- Built numerical algorithms for molecular simulation, optimization, and reaction-path discovery.

PhD Researcher

2018–2023

KU Leuven

- Invented a multiscale Markov chain Monte Carlo algorithm for molecular dynamics.
- Developed theory and high-performance software for discovering new stable protein conformations.

Software Engineer

Summer 2017

Facebook (Meta)

- Integrated a C++ rigid-body physics engine into Facebook's augmented reality platform.
- Enabled physically realistic simulations for Messenger and Instagram.

Selected Side Projects

Autonomous Research Agents

- Designed autonomous AI systems capable of planning numerical experiments, invoking computational tools, evaluating results, and iteratively refining scientific solutions with minimal human supervision.
- Built reusable agent infrastructure, provider-agnostic workflows combining LLM reasoning, structured outputs, tool orchestration, retrieval, and the Model Context Protocol (MCP).
- Developed research agents that automatically optimize PDE solvers, generate hypotheses, select hyperparameters, and produce research-grade analyses for computational science.
- Code can be found on my GitHub page.

Reaction Studio

- Built an AI-native computational chemistry platform featuring diffusion models, conformer generation, geometry optimization, and transition-path search.
- Designed production infrastructure using FastAPI, PostgreSQL, AWS, React, GPU computing, and ML workflows.

PyCont-Lite

- Developed a lightweight numerical continuation package modernizing legacy scientific software for bifurcation analysis and nonlinear dynamical systems.

Education

KU Leuven, Belgium

PhD, Applied Mathematics, specializing in Computational Chemistry

2023

M.Sc., Mathematical Engineering

2018

B.Sc., Computer Science & Electrical Engineering

2016

Technical Skills

Programming	Python • C++ • MATLAB • SQL
AI	LLMs • Agentic AI • Prompt Engineering • Tool Calling • Structured Outputs • Transformers
AI Systems	MCP • RAG • LangGraph • AutoGen • Vector Databases
Scientific Computing	CUDA • HPC • MCMC • Numerical Linear Algebra • Uncertainty Quantification